# FCW: A Forward Collision Warning System Using Convolutional Neural Network

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***Abstract*—In this research paper, our approach is to present a practical solution for all the road car accidents which takes place. Our model will monitor the preceding vehicles and simultaneously it will calculate the distance between the vehicles. Now, if the preceding vehicle gets too close, then our model willwarnthat driver of an impending crash.The driver of the vehicle will simply be alerted by any audible signal after reaching the threshold distance between the vehicles. This model will be very much helpful, especially for the learners. This calculation of the distance between the vehicles will be done by our CNN based model and a camera which will be located on the windshield to click the rear side images of the preceding vehicle in real time. In our experimental results, we can see that our model can calculate the distance and we would be able to achieve an overall good accuracy. In the past, researchers used these technologies like ADAS and FCW system which are entirely dependent on the hardware and cost-effective as well. So, we are using CNN in FCW system instead of using the hardware like radars. In this paper, the concept of Advanced Driver Assistance System (ADAS) and Forward Collision Warning (FCW) are also described.**

***Index Terms*—Convolutional Neural Network; advanced driver assistance system; forward collision warning**

## I. Introduction

According to World Health Organization, 1.25 million death was recorded due to road accident in the year 2013 [1]. Getting a news of road accident is very common nowadays. Road accidents are very grievous, and from the above-mentioned data, we can understand that many innocent people die in road accident every year. However, there are many passive accident safety systems like seat belts, airbag etc. but they help in saving life after the collision or accident has occurred. So, there is a high demand of 'something' that can prevent the cars from accident. Advanced Driver Assistance System (ADAS) is that 'something' that is in very need of the today's road. ADAS is a system which provides drivers a kind of assistance while driving. It helps drivers to detect lane [6], to avoid collision with pedestrians [7], and it also helps the driver to recognize the road traffic signals [9], [12], so that the traffic rules can be followed properly. Driver fatigue [11] is another problem that drivers face while driving, as drowsiness and tiredness felt by a human being is a common thing. This drowsiness can make severe harm sometimes but with ADAS it can also be detected. ADAS also provides assistance while car parking [8]. Advanced Driver Assistance System becomes more helpful for a new learner, as it provides a wide range of help for new drivers.

For a driver while driving, it is very much required to estimate the distance of her car from a car which is in front of her car. This is because a safe distance should be maintained between two cars which are moving along a single line one after another in a road. However, the estimation of this distance is a challenging issue for a learner in driving. Furthermore, in this case, the learner should have the capability of predicting the distance by looking at the car that is in front of her car. If a driver spends much time to estimate such distance between two cars which are moving, it would be difficult for the driver to concentrate on the driving. In addition, an inappropriate prediction of such distance would lead to a collision between two cars, which results in an accident in a road.

Considering aforementioned challenges, in this paper, we design a mechanism, called *Forward Collision Warning system (FCW)*, that warns a driver before the collision is going to occur between two cars. FCW system helps in preventing the collision between two cars while moving. FCW can also avoid a collision at the rear end of the car that is moving in front of our model car. While the distance between two vehicles becomes less than a predefined threshold, a warning sound is generated to notify the driver about the danger which would lead to a collision between two vehicles. This warning includes an audible signal. According to previous research works, a driver assistance system uses infrared laser beam to determine the distance between the cars. The transmitter and the receiver are installed at the hood of the car, which transmit infrared beam and then receive it after getting reflected from the object (i.e., car in front). Thus, calculating the distance between the cars is accomplished [2]. In our model, the distance has been calculated through image processing by using Convolutional Neural Network (CNN). A dataset (i.e., training set and test set) has been generated, which consists of images of the rear end of the car taken from different angles at specific distances. Images have been taken under 4 distances (1.0 metre, 1.5 metres, 2.0 metres, 2.5 metres) and the threshold value is kept at 1 metre. If the distance between the cars is less than the threshold value, an audible signal will be generated and warns the driver about the close distance between the two cars, which can results in a road accident.

The rest of the paper is organized as follows. Section II discusses the related works in ADAS. Section III presents our model description in detail. We mention the results in Section IV. Finally, conclusion is mentioned in Section V.

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## II. Related Works

As we know how the world is changing rapidly, the automobile industry is also changing. The way vehicle works today is entirely different from what it was a decade ago. The rapid change also demands more safer road, so with time there is a high demand of Advanced Driver Assistance System (ADAS) in automobile industry, and whichhas been used extensively in many of the countries (especially in Europe). According to World Health Organization (WHO) [3] 20.7 in a 100000 of population people died in road accident only in Asian continent. A data from the Indian govt [4] is not different at all.

Forward collision warning (FCW) is a preemptive approach to reduce the risk of collision of the vehicle. Many works have been done in this field, the paper [10] is an example of that, in this model monitors drivers risk tolerance. Each driver has his/her own vulnerability level. This model sets a threshold for the drivers risk tolerance level. It compares the instantaneous drivers risk tolerance with the normal drivers risk tolerance level, this parameter of normal drivers risk tolerance is mentioned in the model from the beginning, whenever the instantaneous risk tolerance falls below the threshold risk tolerance it generates a warning alarm for the driver.This model has another part, in which it detects the drivers distraction, here a camera monitors the behavior of the driver and warns the driver about his/her distraction, a CNN model is being trained for this purpose.

Drivers emotion detection [13] is another approach in Advanced Drivers Assistance System (ADAS) which helps in FCW. In this a voice alert system has been setup which generates voice alerts by analyzing the emotion of the driver. Whenever, it detects the positive emotions of driver it generates a moderate alert for a fixed duration, and whenever negative emotion is detected like sad, anger, anxiety, disgust, then it generates a highly elaborative alert to the driver and makes the driver to understand the criticality of driving and significance of safe driving. Emotion of the driver are detected from face recognition, emotion recognition from voice, analyzing body temperature and blood pressure. This model works mainly on facial recognition through a Deep Learning system. In this a Convolutional Neural Network is being developed and trained with the facial emotion data that are feeded into the model.

Fatigue detection [11] also works in ADAS, in which the drivers physical condition and mental conditions are being analyzed and a warning is given to the driver for his/her conditions and sometimes override the brake system of the vehicle to stop the car. This model very accurately analyzes the drowsiness of the driver by monitoring the expression of the face, movement of the eyelid of the driver and lips expression. It keeps a very precise data about the yawn frequency of the driver. This model works on Machine Learning using the Support Vector Machine Algorithm.

TABLE I
Dataset

| Distance | Training set | Test set | Validation set |
|---|---|---|---|
| 1.0 metre | 571 | 50 | 20 |
| 1.5 metres | 560 | 50 | 50 |
| 2.0 metres | 729 | 50 | 20 |
| 2.5 metres | 683 | 50 | 20 |

## III. Model Description

### A. Our Proposed Model

We use CNN in our mechanism and the reason behind choosing this model is as follows. Distance calculation via the infrared transmitter and receiver is a bit complicated process in which every transmitter should have beam of different frequencies and also, the cost is high. But in our case, it is a simple process and also, it is software based with less number of hardware involved and thus, it is cost effective. In our model, we will use a camera to capture the images of the rear side of the preceding car. The images of the car can be captured, from any angle of the camera in real time. After capturing the images of the car, the images will be fed to our model. Next, after feeding those images in our model, it will then detect the distance between the cars. The model will calculate the distance between them using the Convolution Neural Network (CNN) which is described in the Image Processing using CNN section (II - B).

After calculating, if its found that the distance between the cars is less than the threshold distance, then our model will generate an alarm to alert the driver of the car. The intensity of the alert generated, increases as the cargoes further near to the preceding car until it goes beyond the threshold distance.

### B. Image processing using CNN

For our model, firstly we use some image captured by the camera of two cars model(viz. Chevrolet Spark LT and Hyundai i20) to make the dataset for training our model. The dataset consists of 2853 images which are divided into training set, test set and validation set with 2543, 200, 110 images respectively (Table I).

We feed these images into a classier to get the predicted output result. The proposed approach of the image classication is based on the steps as described below.

During the first step, we initialize the Convolution Neural Network (CNN) by defining it as a sequence of layers. Next, we apply the Feature Detector or filter on our dataset. Here we have used 32 feature detectors of $3 \times 3$. Feature detector reduces the size of the input image and due to this some features are lost.

The way we prevent loss of image information is by having many Feature Maps. Each feature map detects the location of certain features in the image. After that we increase the non-linearity in the CNN by applying the rectifier function. Here we have applied ReLu activation function. This function generally removes the negative pixels from the feature maps.



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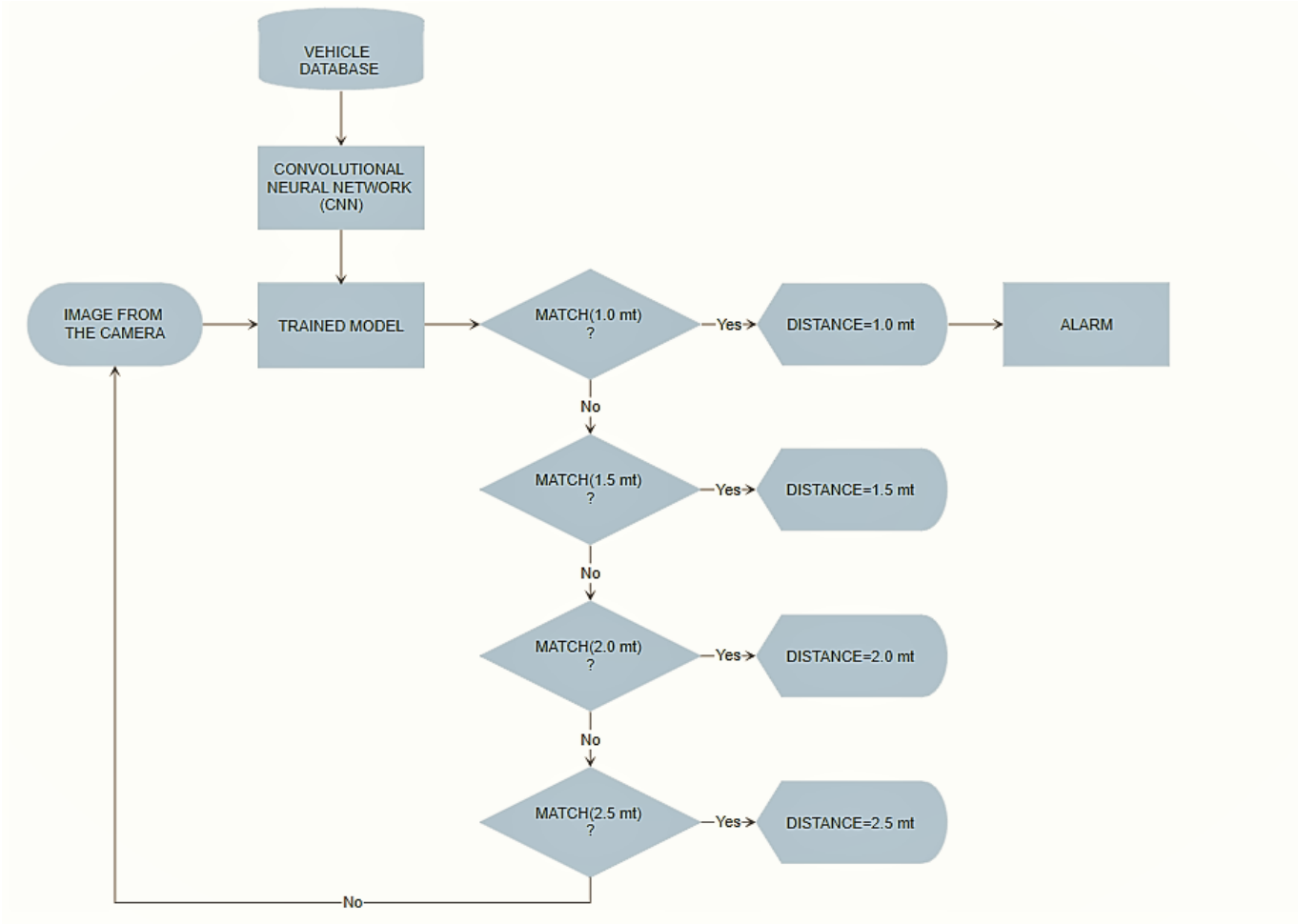


Fig. 1. Flowchart

Then, we add the pooling layer to the feature maps and flattens the pooled images. Then after flattening, the flattened feature map is passed through a neural network. This step is made up of the input layer, the fully connected layer, and the output layer. The size of input image in the first convolutional layer is of $64 \times 64$. The pooling size is of $2 \times 2$ matrix. The filter size is $3 \times 3$ and the number of filters used is 32.

Subsequently, now we have the fully connected layer, which is similar to the hidden layer. Here we have 128 nodes in the hidden layer and 5 nodes in the output layer.We have applied ReLu activation function in the hidden layers and SoftMax activation function in the output layer to acquire the best classification accuracy and also because of more than two outcomes in the output layer. We compile the CNN with the compile function using the Adaptive moment estimation (Adam) asan optimizer and Categorical crossentropy as the loss function and Accuracy as the metrics of performance.

Here we have used Keras to pre-process the images and it also prevents overfitting. This processing is known as image augmentation. The Keras utility we use for this purpose is ImageDataGenerator [5]. This function works by flipping, rescaling, zooming, and shearing the images. In the training set we have used $flow_from_directoryfunction$ and here we have taken 32 as the batch size, 64 as the target size and Categorical as class mode. Batch size is the number of observations after which the weights are updated. Target size is the number of input size pixels and class mode tells us about the number of categories of the dependent variables. Similarly, we have created the test set with the same parameters. After that, we fit the model to the training dataset and test its performance with the test set and then we have used the predict method to make new predictions using new images.

In case of a traffic jam, when there is almost no gap between the cars, the audible signal will be generated continuously inside the car. That might become frustrating, so to tackle this problem, there will be an option to mute the audible signal for a certain time period which can be controlled explicitly by the drivers. However, it is suggested that learners should not use this feature. Here, the mute option will simply disable the audible signal for a certain time period and after that period, the alarm system will automatically start working in a normal way.



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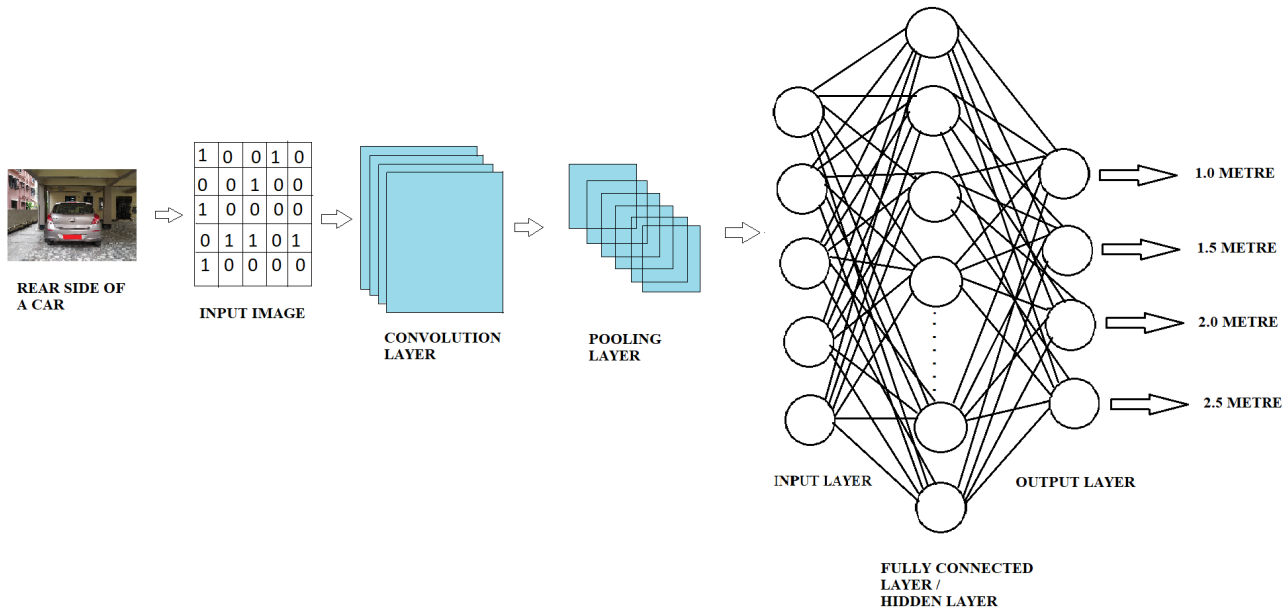


Fig. 2. Performance in terms of throughput

TABLE II
PERFORMANCE ANALYSIS

| Distance | Validation set | Correct prediction | Wrong prediction | Accuracy (%) |
|---|---|---|---|---|
| 1.0 metre | 20 | 14 | 6 | 70 |
| 1.5 metres | 50 | 48 | 2 | 96 |
| 2.0 metres | 20 | 12 | 8 | 60 |
| 2.5 metres | 20 | 17 | 3 | 85 |

## IV. PERFORMANCE ANALYSIS

About 70% of the road accident is due to driver negligence. We can prevent the accident by using this model. Here accuracy is 70%, 96%, 60% and 85% for the distances 1.0 metre, 1.5 metres, 2.0 metres and 2.5 metres respectively (See Table II).

The average accuracy is 77.75% which will increase with increase in number of images in dataset.

## V. CONCLUSION AND FUTURE DIRECTION

Our paper mainly focuses on generating a warning so that a tragic accident can be avoided before it happens, and our proposed model has a good future scope in ADAS for reducing the risk of forward collision. Here we have implemented CNN to create a model to detect the distances between the preceding cars and to maintain a safe distance and if the car somehow crosses the threshold distance then it generates an alarm to warn the driver. There is a chance that the threshold distance for different model of different cars can be different as the front part of the car i.e., the hood can differ in size for every other car. Some of the key features that can be adapted in our model to make it more precise and accurate are that this model

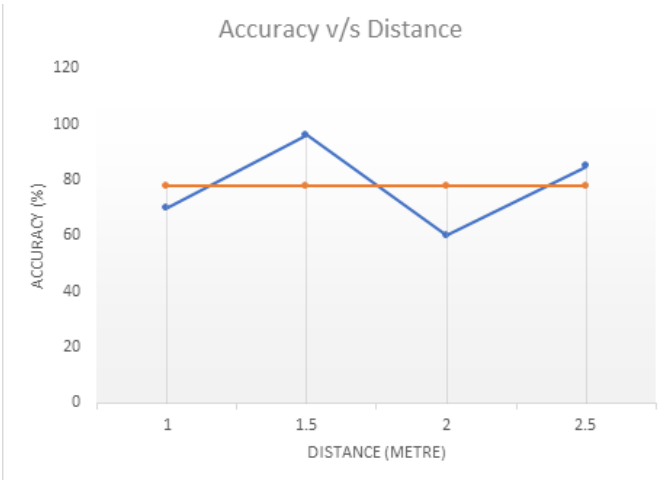


Fig. 3. Performance in terms of throughput



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for now only works on two car models but in future more data of several cars can be added that will make it more robust, and can be smoothly implemented in modern cars to avoid forward collision. We shall use microcontroller for practical implementation of our model which would be described in detail in our future work. We shall also increase the size of the dataset in our future direction of the work. The model would be cost effective as it is software based and it will have quite a less number of hardware (i.e., Microcontroller and a camera).

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